

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Examination, Autumn 2025

Course Code: CSE-4747 Course Title: Mathematical Analysis for Computer Science

Time: 2 Hours 30 Minutes

Total Marks: 50

[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

CLOs DL M

1. A company is reviewing a job applicant. Each interviewer independently gives either a positive or negative recommendation. The review process continues until the applicant receives two positive recommendations or two negative recommendations, whichever happens first. Then a final decision is made. Assume that each interviewer gives a positive recommendation with probability $3/5$, independent of previous interviews.
- a) CLO2 Ap 05

Answer the following using the four-step method:

- What is the probability that three interviews are conducted?
- What is the probability that the applicant is ultimately accepted (two positives) even though the first interviewer gave a negative recommendation?
- What is the probability that the applicant is ultimately accepted?

1. Suppose that we flip three fair, mutually-independent coins and consider the following three events: CLO2 An 05
- b)
- o A1 is the event that coin 1 matches coin 2.
 - o A2 is the event that coin 2 matches coin 3.
 - o A3 is the event that coin 3 matches coin 1.

Now, illustrate that how pairwise independence does not necessarily imply the mutual independence.

OR

1. There is an unpleasant, degenerative disease called Beaver Fever which causes people to tell math jokes unrelentingly in social settings, believing other people will think they're funny. Fortunately, Beaver Fever is rare, afflicting only about 1 in 1000 people. Doctor Islam has a fairly reliable diagnostic test to determine who is going to suffer from this disease: CLO2 An 10

- If a person will suffer from Beaver Fever, the probability that Dr. Islam diagnoses this is 0.99.
- If a person will not suffer from Beaver Fever, the probability that Dr. Islam diagnoses this is 0.97.

Let B be the event that a randomly chosen person will suffer Beaver Fever, and Y be the event that Dr. Islam's diagnosis is "Yes, this person will suffer from Beaver Fever," with B^c and Y^c being the complements of these events.

Now

- The description above explicitly gives the values of the following quantities. What are their values? find $\Pr[B]$, $\Pr[Y|B]$, and $\Pr[Y^c|B^c]$
- Write formulas for $\Pr[B^c]$ and $\Pr[Y|B^c]$ solely in terms of the explicitly given quantities in part (i)—literally use their expressions, not their numeric values.
- Write a formula solely in terms of the expressions given in part (i) for the probability that a person will suffer Beaver Fever given that Doctor Meyer says they will. Then calculate the numerical value of the formula.

2. Three sensors S_1, S_2 and S_3 independently report either ON or OFF with equal probability. CLO2 An 04

Let consider the following events:

- A_1 : Sensors S_1 and S_2 report the same state
- A_2 : Sensors S_2 and S_3 report the same state
- A_3 : Sensors S_1 and S_3 report the same state

Determine whether the events A_1, A_2 , and A_3 pairwise independent, and mutually independent.

2. Some search algorithms randomly explore a solution space until they find a desired outcome (success). Consider a maze-solving algorithm where each move has a 20% chance of leading to the exit (success) and 80% chance of hitting a wall (failure). CLO2 Ap 03

- (i) What is the probability of finding the exit within 10 attempted moves?
- (ii) How many moves would you expect (on average) before reaching the exit?

2. During the testing phase of a new software, bugs are discovered at an average rate of 0.5 bugs per hour. CLO2 Ap 03

- (i) What is the probability of finding no bugs in the next 2 hours of testing?
- (ii) If the testing team aims to find at least 2 bugs before releasing the software, what is the probability they will need more than 4 hours of testing?

Group B

3. Explain Simpson's paradox. CLO2 An 05
- a) A company is hiring for two teams: Engineering and Marketing. The data for applications and hires is as follows:

Team	Male Applicants	Male Hired	Female Applicants	Female Hired
Engineering	80	56	20	16
Marketing	20	12	80	24

- i. Calculate the hiring rate for males and females within each team.
 - ii. Calculate the overall hiring rate for males and females (across both teams).
3. Suppose a computer program has a probability p of crashing out at the end of each hour of use. Now formulate a procedure to estimate the expected time until the computer program crashes out? CLO2 Ap 05

4. Random walks are commonly used to estimate the relative importance (ranking) of web pages in a network, as in Google's PageRank algorithm. The basic idea is that a random surfer navigates the web by randomly following outgoing hyperlinks from one page to another. CLO3 E 05
- a)

Consider the following Web-graph consisting of four web pages:

- Page P links to Page Q
- Page Q links to Page R and Page S
- Page R links to Page P
- Page S links to Page R

- i. Represent the given Web-graph using a diagram, where web pages are shown as nodes and hyperlinks are shown as directed edges.
- ii. Explain how a random walk can be performed on this Web-graph to estimate the relative rank of each page.
- iii. Describe how the probability of the random surfer being on each page evolves over time and explain why these probabilities stabilize after a large number of steps.

OR

4. You want to analyze user navigation patterns on an e-commerce website with three main sections: CLO3 An 05

a) Products (P): Browse product listings.
 Cart (C): View items added to the cart.
 Checkout (O): Process the order and payment.
 You collected data on user transitions between these sections and estimated the following transition probabilities:

Transition From	Products (P)	Cart (C)	Checkout (O)
Products (P)	0.6	0.3	0.1
Cart (C)	0.2	0.5	0.3
Checkout (O)	-	-	1

- (i) A user starts browsing products (State P). Simulate the next two transitions (three states total) using the transition probabilities. What is the probability of the user reaching the checkout (State O) within these two transitions?
- (ii) In the long run, what is the probability of a user being in each section (Products, Cart, Checkout)?
4. A university tracks the choice of specialization made by students across academic years. CLO3 E 05
- b) Each year, a student may either continue in the same specialization or switch to another. The three specializations offered are: D(Data Science), R(Robotics) and I(Information Systems).

The transition probabilities from one academic year to the next are defined as follows:

- A student currently enrolled in Data Science (D) remains in Data Science with probability 0.6, switches to Robotics with probability 0.3, and switches to Information Systems with probability 0.1.
- A student currently enrolled in Robotics (R) switches to Data Science with probability 0.3, remains in Robotics with probability 0.6, and switches to Information Systems with probability 0.1.
- A student currently enrolled in Information Systems (I) switches to Data Science with probability 0.2, switches to Robotics with probability 0.2, and remains in Information Systems with probability 0.6.

Assume that specialization choices are reviewed once every academic year, so the transition period is one year.

Questions

- (i) Find the probability that a student who is enrolled in Data Science (D) in the current academic year will be enrolled in Robotics (R) in the next academic year.
- (ii) Find the probability that a student who is enrolled in Data Science (D) in the current academic year will be enrolled in Robotics (R) two academic years from now.
5. State Kendall notation with examples. Derive the steady state probability of M/M/1 queuing system. CLO3 An 05
- a) system.
5. A call center operates with an M/M/1 system. Calls arrive at a rate of $\lambda = 5$ calls per minute. CLO3 E 05
- b) The average call duration (including agent handling time) is $\mu = 4$ minutes. The call center aims to maintain an average waiting time of no more than 2 minutes.

- (i) Current Performance: Calculate the current utilization factor, average queue length, and average waiting time for the call center.
- (ii) Staffing: Based on the target waiting time, determine the minimum number of agents required if each agent can handle one call at a time.
- (iii) Impact of Voicemail: How would offering a voicemail option for callers affect the system performance? Analyze the impact on queue length and waiting time.

OR

5. A popular food delivery app experiences high order volume during lunch hours. Their system processes orders in an M/M/1 fashion. Orders arrive at a rate of $\lambda = 20$ orders per minute on average. The average processing time per order (including restaurant confirmation and delivery assignment) is $\mu = 1.5$ minutes. CLO3 E 05

- (i) Customer Satisfaction: Calculate the current utilization factor, average queue length, and average waiting time for an order to be processed. Analyze if this might lead to customer dissatisfaction with long wait times.
- (ii) Dynamic Pricing: The app considers implementing dynamic pricing based on demand. During peak hours, the price might increase slightly to discourage a surge in orders and maintain a manageable workload. How would such a strategy impact the arrival rate (λ), and consequently, the waiting times for customers who are willing to pay the higher price?
- (iii) Delivery Partner Matching: Optimizing delivery partner assignments can improve overall processing efficiency. How can the app leverage queuing theory concepts to improve delivery partner matching and potentially reduce order processing time (μ)?

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Term Examination, Autumn-2025

Course Code: CSE 4741

Course Title: Computer Graphics

Total marks: 50

Time: 2 hours 30 minutes

[Answer all the questions; Separate answer scripts must be used for Group A and Group B]

GROUP A

- | | M | CO | DL |
|--|---|-----|----|
| 1. a) What is viewing transformation? Find the general transformation matrix for window-to-viewport mapping. | 4 | CO2 | C2 |
| 1. b) Find the region codes for end point A(5,2); B(3,8); C(-1,2), D(2,4) and E(7,8) & F(3,5). Then, Clip the line segments AB, CD and EF using Cohen-Sutherland algorithm. (Assume $X_{min}=-1$, $X_{max}=5$, $Y_{min}=-1$, $Y_{max}=5$). | 6 | CO3 | C2 |
| OR | | | |
| 1. b) Find the region codes for end point A(5,2); B(3,8); C(-1,2), D(2,4) and E(7,8) & F(3,5). Then, Clip the line segments AB, CD and EF using Liang-Barsky algorithm. (Assume $X_{min}=-1$, $X_{max}=5$, $Y_{min}=-1$, $Y_{max}=5$). | 6 | CO3 | C2 |
| 2.a) Define tilting as a rotation about the x axis is 30° followed by a rotation about the y axis is 45° . a) Find out the rotation matrix. b) Does the order of performing the rotation matter? | 4 | CO4 | C3 |
| OR | | | |
| 2.a) What are the advantages of the Cohen-Sutherland algorithm over Sutherland-Hodgman algorithm for clipping? Show the comparison between geometric and coordinate transformation. | 4 | CO4 | C3 |
| 2.b) Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY, YZ, ZX plane and find out the new coordinates of the object. | 4 | CO1 | C2 |
| 2.c) Write the differences between 2D rotation and 3D rotation. | 2 | CO1 | C2 |

GROUP B

- | | | | |
|--|---|-----|----|
| 3. a) Given a 3D point P(2,5,10), the scene is rendered using perspective projection with the camera at the origin and the viewing plane at $z=5$. Calculate the 2D projected coordinates of the point P on the viewing plane. Also, explain how the z-coordinate (depth) affects the size of the projection. | 6 | CO1 | C2 |
| 3.b) The unit cube is projected onto the xy plane. Draw the projected image using the standard perspective transformation with
(i) $s = 4$
(ii) $s = 15$
where s is distance from the view plane. | 4 | CO2 | C3 |
| 4.a) Explain the Z-buffer algorithm for hidden surface removal in 3D rendering. How does the algorithm determine which surfaces are visible in a scene? Describe the | 5 | CO2 | C3 |

steps involved in the Z-buffer algorithm, and discuss its advantages and limitations compared to other hidden surface removal techniques like the Painter's Algorithm.

OR

- | | | | | |
|------|---|---|-----|----|
| 4.a) | "Phong model is widely used to mimic the reflection of light from object surfaces to the viewer's eye". Describe the Phong model in brief. | 5 | CO2 | C1 |
| 4.b) | In the Painter's Algorithm, objects are painted based on their distance from the viewer. Given three objects in a 3D scene with coordinates A(1, 1, 2), B(2, 3, 5), and C(4, 4, 8), explain how the algorithm would order these objects for rendering. What challenges can arise when using the Painter's Algorithm, and how can they be addressed? | 5 | CO2 | C3 |

OR

- | | | | | |
|------|--|---|-----|----|
| 4.b) | Give a suitable example of Scan Line Hidden Surface Removal algorithm and describe it. | 5 | CO2 | C3 |
| 5.a) | Describe a recursive ray tracing method. How ray tracing algorithms help to represent realistic pictures? | 4 | CO2 | C1 |
| 5.b) | What is light? What are the basic characteristics of light? Describe the visible band of electromagnetic spectrum. | 4 | CO2 | C1 |
| 5.c) | A ray is represented by $\vec{r}(t) = \vec{s} + t\vec{d}$ and $\vec{d} = 2\vec{i} + 3\vec{k}$. Find the coordinates of the points on the ray that corresponds to $t=2, 3$ and 4 respectively. | 2 | CO2 | C3 |

$$r(t) = s + td$$

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in Computer Science and Engineering

Semester Final Examination, Autumn 2025

Course Title: Machine Learning and Data Mining

Course Code: CSE-4877

Full Marks: 50

Time: 2 Hours 30 Minutes

[Answer all the questions. Figures in the right margin indicate full marks. Separate answer scripts must be used for Group A and Group B]

Group A

1 a) Identify the machine learning task and justify your answer:

- An online store segments customers based on purchasing behavior.
- A company predicts the salary of an employee based on experience and education.
- A social media platform identifies whether a post contains hate speech.

Given the following training data:

Age	Income	Student	Credit Rating	Buys Computer
≤ 30	High	No	Fair	No
≤ 30	High	No	Excellent	No
31-40	High	No	Fair	Yes
> 40	Medium	No	Fair	Yes
> 40	Low	Yes	Fair	Yes

Calculate the Information Gain for the attribute Age.

c) The following table shows the number of hours studied and the corresponding exam scores of students:

Hours Studied (X)	Score (Y)
2	50
4	60
6	65
8	80
10	85

- Using simple linear regression, compute the regression equation $Y=a+bX$.
- Predict the expected score of a student who studies for 7 hours.
- Briefly comment on the relationship between study hours and score

2 b) A binary classification system is tested on 10,000 medical samples. The performance results are summarized in the following table:

Actual \ Predicted	Positive	Negative	Total
Positive	90	210	300
Negative	140	9560	9700
Total	230	9770	10000

- Compute the performance measures: Accuracy, Precision, and Recall.
- Based on your calculated values, critically evaluate whether this model is reliable for detecting positive cases.
- From the calculated metrics, identify one measure that best reflects the model's true detection capability and justify your answer.

OR

Suppose that we want to select between two prediction models, M1 and M2. We have performed 10 rounds of 10-fold cross-validation on each model, where the same data partitioning in round i is used for both M1 and M2. The error rates obtained for M1 are:

30.5, 32.2, 20.7, 20.6, 31.0, 41.0, 27.7, 26.0, 21.5, 26.0

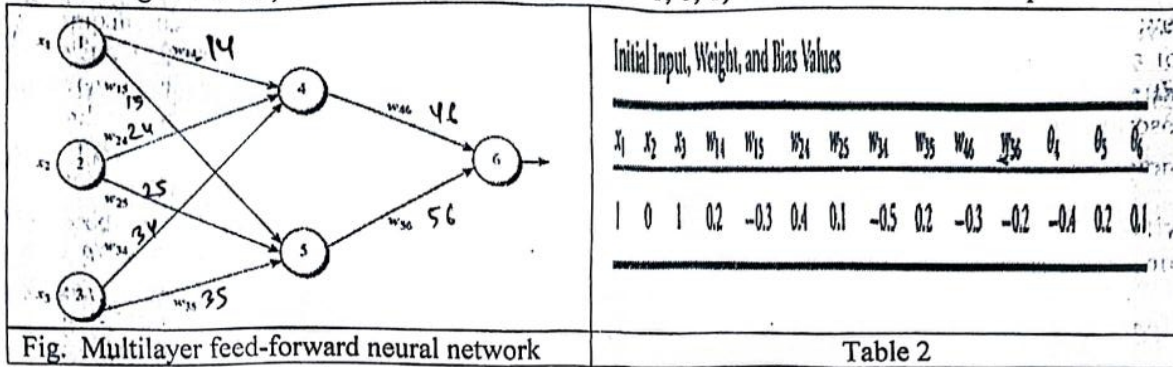
The error rates for M2 are:

22.4, 14.5, 22.4, 19.6, 20.7, 20.4, 22.1, 19.4, 16.2, 35.0

Comment on whether one model is significantly better than the other considering a significance level of 1%.

Group B

- 3 a) The given multilayer feed-forward neural network has specific initial weights and biases (Table 2). The learning rate is 0.9, and the first training tuple ($X = 1, 0, 1$) with a class label of 1 is provided.



Answer the following questions:

- Show net Input and Output Calculations without transfer function.
 - Show net Input and Output Calculations with relevant transfer function.
 - Show the calculations for weight and bias updating.
- b) Answer the following questions with respect to neural network:
- If the error becomes very small during training, how does it affect weight updates and what does this indicate about the learning of the neural network?
 - Explain how the choice of learning rate affects the training behavior and convergence of a neural network. Why is it important to avoid both very small and very large learning rates?

OR

- b) A company uses multiple different models to decide whether a loan application should be approved or rejected. During testing, it is observed that a single model often gives unstable predictions for different applicants. As a machine learning engineer, propose a solution strategy to improve the reliability and robustness of the decision-making system. Justify, why your proposed approach would perform better.

- 4 a) Suppose a binary classification problem has the following data points: Positive class: $\{(2,2), (2,0), (3,2), (3,0)\}$, Negative class: $\{(6,1), (7,2), (7,0), (8,1)\}$. The support vectors are: $\{(3,2), (3,0), (6,1)\}$
- Sketch the data points and the separating hyperplane.
 - Determine the equation of the separating hyperplane.
 - Classify the data point $(6,3)$ using SVM.

- b) What factors should be considered when choosing the value of 'k' in the k-Nearest Neighbors (k-NN) algorithm, and how do these factors impact the model's performance in terms of bias-variance tradeoff, class imbalance, and computational efficiency?

OR

- b) Discuss and analyze the fundamental differences between K-means clustering and K-Nearest Neighbors (KNN) classification, emphasizing their learning objectives, algorithmic behavior, and practical limitations.

- 5 a) Identify the type of outlier and justify your answer.
- Most students attend classes 4–5 days per week, but one student attends only 1 day every week throughout the semester.
 - The temperature in a city during winter months is usually 10–15°C, but on one winter day the temperature is recorded as 35°C.
 - A patient's heart rate is normal most of the time, but a short sequence of abnormal readings appears together.

- b) Given the two-dimensional data points : $(4.5, 8), (5, 7), (6, 6.5), (7, 5), (9, 4), (7, 3), (8, 3.5), (9, 5), (4, 4), (3, 7.5), (4, 6), (3.5, 5)$
Identify outlier using a density based clustering algorithm with $\epsilon = 1.9$ and Minimum Points=5.
- c) Explain why separating data into training and testing sets is essential in machine learning

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in Computer Science and Engineering

Semester Final Examination, Autumn 2025

Course Title: Computer Networks

Course Code: CSE-3633

Full Marks: 50

Time: 2 Hours 30 Minutes

[Answer all the questions. Figures in the right margin indicate full marks. Separate answer scripts must be used for Group A and Group B]

Group A

1. a) How is queue length used to control congestion with Random Early Detection algorithm? Analyze with suitable figure. DL An CO 3 5 ✓

OR

- a) Discuss significance of any 3 fields of IPV6 header. Ev CO2 5
b) How does quality of service requirement vary for different applications? Discuss with example. An CO2 5 ✓
2. a) Differentiate between open loop and closed loop congestion control mechanism? Explain why the Hop-by-Hop Choke Packet congestion control mechanism outperforms the simple End-to-End Choke Packet mechanism with a neat and labeled diagram. An CO2 5
b) Write down the advantages and disadvantages of hierarchical routing. Support your answer with an appropriate network diagram to illustrate how hierarchical routing works in a multi-level network. An CO2 5

OR

- b) Computer B has 25 MB of data to send on a network and transmits the data in a burst at 8 Mbps. The maximum transmission rate across routers in the network is 5 Mbps. If Computer B's transmission is shaped using a leaky bucket, how much capacity must the queue in the bucket hold to avoid discarding any data?

Group B

3. a) Analyze the differences between TCP and UDP in terms of their header fields. An CO2 5
b) Explain TCP three-way handshaking process for connection establishment. U CO3 5
4. a) With a diagram, show how buffers are allocated dynamically by a receiver for a sender. Also, show a case in which the sender is deadlocked, what should be done to prevent such deadlocks? An CO1 5
b) Explain how TCP implements flow control using the sliding window mechanism. A TCP receiver has a receive window of 8000 bytes. The sender has sent 5000 bytes, and 2000 bytes have been acknowledged. What is the current window size available to the sender? An CO4 5
5. a) Suppose you want to access the website "iiuc.ac.bd" and your local DNS server does not have its IP cached. Explain how the IP address is resolved using the DNS iterative method? Un CO3 5
b) What is a CDN (Content Delivery Network)? Explain how CDNs work with figures and why they are important for modern web applications. Un CO3 5

OR

5. a) The students and teachers of IIUC use Telegram to communicate and share information. They create groups and channels to share class announcements, lecture notes, assignments, and project updates. Some messages are text, others include files, images, and videos. Messages are delivered in real-time, and some require end-to-end encryption for privacy. Occasionally, multiple students send messages or upload files at the same time.
i. Which network architecture (Client-Server or Peer-to-Peer) is used by Telegram? Explain briefly with proper figures. ✓
ii. Identify advantages and disadvantages of the chosen architecture.
b) What is the key advantage of IMAP over POP3? Explain how SMTP, POP3, and IMAP protocols work together in the process of sending and receiving emails?

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in Computer Science and Engineering

Semester Final Examination, Autumn 2025

Course Title: Numerical Methods

Course Code: CSE-4745

Time: 2 Hours 30 Minutes

Full Marks: 50

[Answer all the questions. Figures in the right margin indicate full marks. Separate answer scripts must be used for Group A and Group B]

Group A

- | | DL | CO | Marks |
|--|----|-----|-------|
| 1. a) Solve the following system of linear equations using the Basic Gauss Elimination method:
$3x + y + z = 10$
$2x + 4y + z = 12$
$x + y + 5z = 14$
OR
Solve the following system of linear equations using Jacobi's iterative method. Carry out four iterations, starting with the initial approximation $x = 0, y = 0, z = 0$.
$10x + 2y + z = 9$
$2x + 10y + 3z = 13$
$x + 2y + 10z = 14$ | A | CO2 | 4 |
| b) Describe the matrix inversion method used to solve a system of linear equations. | U | CO1 | 4 |
| c) Gauss-Seidel method is similar to Jacobi's method for solving systems of linear equations. State the differences between the Gauss-Seidel method and Jacobi's method. | U | CO1 | 2 |
| 2. a) What do you mean by curve fitting? Compare curve fitting with interpolation. | U | CO1 | 2 |
| b) Describe the least square method to fit a straight line. | U | CO1 | 4 |
| OR
Describe the least square method to fit a nth degree polynomial. | | | |
| c) In some determination of the volume v of carbondioxide dissolved in a given volume of water at different temperatures θ , the following pairs of values were obtained: | A | CO2 | 4 |

θ	0	5	10	15	20
v	1.70	1.35	1.00	0.60	0.20

Find a relation of the form $v = a + b\theta$, which best fits to these observations by the method of least squares.

Group B

- | | | | |
|---|---|-----|---|
| 3. a) Determine the first order and second order derivatives using Newton's forward interpolation formula. | U | CO3 | 4 |
| b) A rod is rotating in a plane. The following table gives the angle θ in radians through which the rod has turned for various values of the time t seconds. Calculate the angular velocity and the angular acceleration of the rod, when $t = 0.3$ sec. | A | CO4 | 4 |

t seconds	0.0	0.2	0.4	0.6	0.8	1.0
θ radians	0.0	0.32	0.58	0.95	1.52	2.28

- | | | | |
|--|---|-----|---|
| c) Explain how numerical differentiation can be used to find the maximum and minimum values of a tabulated function. | U | CO3 | 2 |
|--|---|-----|---|

4. a) Derive Simpson's 1/3 rule for numerical integration using Newton's forward interpolation formula. U CO5 5

OR

Derive Simpson's 3/8 rule for numerical integration using Newton's forward interpolation formula.

- b) A Civil engineering team is designing a road embankment across a low-lying area. To estimate the amount of earth required, the height of the embankment (in meters) is measured at equal distances along a straight line. A CO6 4

The recorded data are given below:

Distance (m)	0	5	10	15	20	25	30
Height (m)	0	1.8	3.6	4.2	3.9	2.1	0

Using the above data, estimate the area of the cross-section of the embankment by applying:

- (i) Simpson's 1/3 rule
(ii) Simpson's 3/8 rule

- c) Write the formula of Weddle's rule for numerical integration

R CO5 1

5. a) Write the advantages and limitations of i) *Taylor's series method* ii) *Euler's method*. U CO3 2

- b) Describe the *Taylor's series method* for the numerical solution of ordinary differential equation. U CO3 3

OR

Describe the *Heun's method* for the numerical solution of ordinary differential equation.

- c) Write the *classical fourth order Runge-Kutta method* formula. Using this formula, solve A CO4 5

$$\frac{dy}{dx} = x^2 + 2y^2$$

$$y(0) = 1.$$

for the interval $0 < x \leq 0.4$, with $h = 0.2$.

International Islamic University Chittagong

Center for General Education (CGED)

Semester End Examination: Autumn 2025

Program: Undergraduate

Course Code: URIH-4701

Course Title: A Survey of Islamic History & Culture

Time: 2 hours and 30 minutes.

Full Marks: 50

Instructions:

- All Questions are Compulsory.
- Figures in the right margin indicate full marks.
- Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.

Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	Analyze the credit and achievement of <i>Muawia bin Abu Sufian</i> (R.) highlighting his foreign policy and contribution.	10	U	CLO2
2	Estimate the administrative reforms of <i>Abdul Malik bin Marwan</i> in the history of Islam. How did these reforms contribute to world civilization?	10	An	CLO3
3	"Mamun's reign was unquestionably the most brilliant and glorious of all the history of Islam"- Estimate the statement with pragmatic reasons. Or ✓ Estimate the reign of <i>khalifah Harun al-Rashid</i> as a great ruler in the history of the world.	10	An	CLO3
4	Evaluate the contributions of Muslim scholars to the development of mathematics and medicine. How has modern civilization been benefited from these scientific contributions?	10	E	CLO3
5	Write short notes on any two: a) Conquest of Spain in 710 A.D. b) Reforms of <i>khalifah Umar bin Abdul Aziz</i> c) Glory of the City of Baghdad	5 5	E	CLO3